AHMET ULU

Embedded Systems & PCB Design · Electrical and Electronics Engineering Student

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SUMMARY

Electrical and electronics engineering student who builds hardware that works in the field. Embedded systems developer at MUSTECH (smart agriculture) and freelance PCB designer on Bionluk. Comfortable across the whole embedded stack: KiCad schematics and PCB layout, C/C++ firmware on ESP32 and STM32, sensor interfacing, and the Python and web software that moves the data. Seeking an embedded systems or hardware engineering internship.

8 client PCB designs delivered | 5.0/5 Bionluk rating (6/6 five-star) | 14 electronics calculators published

EXPERIENCE

Embedded Systems Developer — MUSTECH (Mustech Agro), Kayseri *Feb 2026 – Present*

- Develop embedded firmware and hardware interfaces for IoT environmental monitoring systems used in smart agriculture.
- Design low-noise sensor interface circuits and integrate wireless telemetry for reliable field data acquisition.
- Build data pipelines that carry readings from distributed sensor nodes to cloud databases for real-time monitoring.
- Designed and built the company website (mustechagro.com.tr) in plain HTML/CSS/JavaScript: bilingual, with weather-based field tools (OpenWeather API) and an FAO-56 irrigation calculator.

Freelance PCB Designer — Bionluk (freelance marketplace), Remote *2025 – Present*

- Delivered 8 custom PCB orders end to end in KiCad, from requirements and schematic capture to layout, 3D review and production-ready fabrication files; 5.0/5 rating with 6/6 five-star reviews.
- **Traffic signal controller**: NodeMCU-based board with a 7-sheet hierarchical schematic, 16 outputs from daisy-chained 74HC595 shift registers and PCF8574 I/O expansion.
- **Peltier control board**: ESP32-WROOM-32 power and control board driving six Peltier modules with PWM, plus fans, pumps and sensors (9-sheet schematic).

SELECTED PROJECTS

Solar-Powered Water Quality Monitor — ESP32 · KiCad · team project *2025*

- Off-grid field station measuring pH, TDS and temperature (DS18B20) with an ESP32; sends alerts to a mobile app when readings leave safe limits.
- Designed and hand-routed my first PCB in KiCad; sized a 24 W solar panel and 12 V 24 Ah battery for continuous operation; worked through pH calibration and sensor noise.

EV Charging Station Data Infrastructure — Raspberry Pi · Python · PostgreSQL · Flask · team project *2024*

- Built the data backbone on a Raspberry Pi: PostgreSQL schema, Arduino-to-Pi serial logging every 2 s, camera-based licence plate recognition (OCR) that opens a charging session per vehicle, and a Flask REST API used by the team's mobile app.

Planner — Smart Planner App — JavaScript (PWA) · Google Apps Script · Telegram Bot API · Gemini API *2026*

- Serverless planner on Google Apps Script and Sheets: Telegram morning briefings and reminders, Gemini-generated day plans, Turkish natural-language quick add and habit streaks; installs on phone and desktop.

Details, photos and write-ups: ahmetulu.com.tr

EDUCATION

B.Sc. in Electrical and Electronics Engineering — Abdullah Gül University, Kayseri *2024 – 2029 (expected)*

Intensive English Program (CEFR B2) — English Path, Dubai, UAE *Jun – Aug 2025*

TECHNICAL SKILLS

Embedded: C, C++, ESP32, STM32, Arduino, Raspberry Pi, I2C, SPI, UART, LoRa, PWM, sensor interfacing and calibration

PCB & Hardware: KiCad (hierarchical schematics, PCB layout, 3D review, Gerber output), prototyping, soldering, multimeter debugging, buck converters

Software: Python, Flask, PostgreSQL, REST APIs, JavaScript, HTML/CSS, Google Apps Script, Git/GitHub, Linux

Tools: MATLAB & Simulink

CERTIFICATIONS

From Schematic to PCB: Layout Fundamentals — Coursera (May 2026) • Introduction to the Internet of Things and Embedded Systems — Coursera • MATLAB Onramp — MathWorks • LIFT UP — Industry-Focused Undergraduate Capstone Projects Conference

LANGUAGES

Turkish (native) · English (B2, professional working proficiency)